L8.5: Orthogonal transformations and rotations

Transcript Language

English

Hello and welcome to the batch two component of the online BSc program and data science

and programming. This is a video on orthogonal transformations and rotations. So, before I start

the video, I want to sort of emphasize that this is maybe not an extremely important video from the

perspective of the data science course. But it is a useful video from a if you find

mathematics interesting and certainly the idea of rotations an[...] on which we are very close to given the material that we have covered so far, are interesting even to people who do

not particularly enjoy mathematics. So, in some sense this video is for a general view

of what kind of mathematics you can study using linear algebra which we have studied so far.

So, let us talk about orthogonal transformations and rotations.

So, let V be an inner product space and T be are linear transformation from V to V.

So, T said to be an orthogonal transformation, if the inner product of Tv and Tw is the same as

the inner product of v and w for all v and w and in the vector space V. So, for any two vectors,

you change them, you transform them using a linear transformation T, then the changed the vectors

maintain the inner product. So, the inner product does not change

after the transformation, this is when you will say that this transformation is an

orthogonal transformation. So, what does it mean for Rn, this is a main sort of intuition

behind why we study orthogonal transformations. So, in these Rn with the usual inner product, a linear transformation T Rn to Rn is orthogonal, if it preserves angles and links.

So, let us understand where this is coming from. So, note that the inner product of

two vectors in Rn, the usual inner product, meaning that dot product

is the norm of v times the norm of w times cosine of the angle between these two

vectors. So, now if you take these two sides, you get Tv dot Tw is inner product of Tv sorry,

norm of Tv times norm of Tw or we can say the length of Tv times the length of Tw

times cosine of the angle between Tv and Tw. And on the other side, we have the length of v

times the length of w times the cosine of the angle between v and w. So, now if the

linear transformation preserves, if these two are equal,

so first of all, and they are equal, remember for all v and w, so in particular, you can take

v to be equal to w in that case, the two sides will give you norm of Tv squared is equal to

norm of v squared norms are always positive, that is why norm of Tv is equal to norm of T.

So, in other words, the lengths are the same. After you apply T, the length of v remains

unchanged. So, length of Tv is the same as length of V. So, that means it preserves length.

And now you apply it to the general situation of v and w. So, you have length of v times length

of w times the cosine of the angle between them, the length of Tv times length of Tw

times cosine of the angle between but length of Tv is the same as length of v length of Tw

is the same as length of w. And that is why on both sides, you get only that the angle between

Tv and Tw is equal to the angle between out cosine of the angle between Tv and Tw is the same as

cosine of the angle between a v and w. And that tells you that the angle between v

and w is the same as the angle between our Tv and Tw. So, it also preserves angles.

That is why we have this this comment, so in other words, orthogonal transformations are

some very useful or interesting. They have some interesting geometry, when we consider the usual inner product, namely the dot product. Here is a fact it is enough in the case of Rn

to demand that the linear transformation preserves length, so in that case, angles automatically get

preserved. So, what we are saying is, since this is for all v and w. So, if you take 3

points and then after transformation they become

like this, so the relative lengths are the same. That means this length is also the same.

And that will sort of force that the angle does not change. So, this is the idea.

I am not proving this. But it is a fact that I think is very visually appealing. So, this is

something you may be able to try and see. Let us see a very nice example of the,

of this inaction. So, this is a video created by our team. So, you look at the

xy plane. And let us look at this T. So, the T here is the identity. So, u is T times v,

v is the vector in yellow. And if you apply T, let us see how it changes. So, if your

T is 0.5 0.87 etc. So, look at how your T is changing, and the u is changing accordingly.

So, when it is minus 1, on the diagonal, it is the opposite vector.

And then it comes back closer to v. And now you are back. So, you can see that what happened here?

These various different T's, maybe I let us play this again. So, if you play it again,

let us look at the T's again. So, u is T times v, when T is the identity matrix,

u is just equal to v, when u changes to 0.5 minus 0.87 0.87 0.5, you get changed. So,

you can see that u is being rotated. So, the length is remaining the same.

So, it is rotating on the same circle that tells you that the length is remaining the same.

So, the point here is that these T s that we have all of them preserve length. And that is why

they are orthogonal transformations. So, these are examples of orthogonal transformations.

So, this is the next slide, finding the rotation matrix in the R2. So, what we saw were examples of

what we call rotation matrices. So, they rotate vectors. So, how do we find rotation matrices

in R2, so consider the standard basis, e1 e2 rotate the plane by an angle theta, that is what we saw in the video. So, it was rotated by those various angles. And corresponding to that angle

that you rotated, there was a T there. So, we want to understand how to get

the matrix corresponding to the rotation by theta. So, if you rotate, then let us say you rotate by

theta, then you get these vectors in red. So, you drop the you drop perpendiculars to these

to the axis so that we can write down what are the coordinates of the vector. And

what you can see is, so if this is theta, then the opposite side, well, it is the hypotenuse.

Since it is rotated, the hypotenuse is the vector in red. So, rotation will not change

the length, so the hypotenuse has length 1, this red thing has length 1.

And this as a result has length sine theta, and this as a result as length cosine theta.

So, the coordinate of this point, or the corresponding vector to this red

line is cosine theta, because that is the length that you get on the X axis when you drop the perpendicular or project and on the Y axis, its sine theta.

Similarly, if you do it for e2 the coordinate you get are minus sine theta and cosine of theta.

So, now we know how to write down the corresponding matrix.

Let T theta we are corresponding linear transformation

that is a linear transformation that you obtain by rotating by theta, then we have seen T theta of 1,

0 is cosine theta sine theta and T theta of 0, 1 is minus sine theta cosine theta.

So, now if you want to express this as a matrix meaning with respect to the standard

ordered basis, then we just saw how to do it. Namely, you take these coordinates for 1, 0

and put it in your first column, take these coordinates for

the image of 01 and put it in your second column. This we have seen in our previous videos.

And notice here that R theta transpose is R minus theta, and R theta transpose times R theta

R theta times R theta transposes identity. And note which we have seen earlier that then that since angles and length. Now, there is a typo here, since angles and length are preserved,

and the standard basis orthonormal, the rotated vectors are also all orthonormal. Why? Why is that

the case? Because the angle between these two vectors cosine theta sine theta and minus sine

theta, cosine theta is still 90 degrees. Because we just rotated.

So, they are orthogonal to each other, and the rotation will not change length. So,

1, 0 and 0, 1 have length 1. So, these two basis vectors also have length 1.

So, there you learn orthonormal basis of R2. So, in other words, these columns that we have here,

for R theta is an orthonormal basis to something what thing and it is worth observing that

that fact is exactly borne out by this identity here, the R theta transpose times R theta is R

theta times R theta transpose is identity. So, I encourage you to check that.

By the way, why is R theta transpose equal to R minus theta. So, if you write down R minus theta,

you get cosine of minus theta, but well, that same as cosine of theta, sine of minus theta, which is minus sine of theta. And then

minus of sine theta, with sine of, minus of sine of minus theta, which is minus

minus sine theta which is just sine theta, and then again, cosine of theta.

So, this is exactly R theta transpose. And now you can check that

R theta transpose times R theta is R theta times R theta transpose identity. So, the point here is

when you apply R theta, you are rotating by theta, when you apply R theta transpose, we just we have

seen that that is the same as applying R minus theta, which is the same as saying that you are

rotating back by theta, you are rotating by minus theta. So, rotate by theta rotate backed

by minus theta, you will exactly get the identity transformation. That is what this is seen.

So, as I said, this video is largely a fun video more to do with

the geometry behind what is all going on. So, let us look at rotations is R3. So, the rotations

about the axes in R3 is the general rotations are a bit more to this difficult to describe.

Since, these clearly preserves angles and distances and our linear transformations,

if you rotate about the axis, then suddenly this is a linear transformation.

And it is clearly want to preserve lengths and angles, this was what we also saw in R2.

So, these are orthogonal transformations. So, in the previous slide, we saw for R2, you could

describe them by looking at the images of e1 and e2 here, you have to look at the images of e1,

e2, e3. So, if we consider the rotation about the Z axis, and then e3 is unchanged, because if you

look at the Z axis that you are rotating about the Z axis, so the Z axis remains unchanged.

So e3, which is a unit vector in the Z axis direction, remains unchanged.

And the xy plane gets rotated exactly as in the previous

slide. So, what we get is the corresponding matrix looks like cosine theta minus sine theta 0 sine

theta cosine theta 001. So, this is because we are basically rotating only the xy plane.

And if you followed that, then the same idea will tell you that if you have the X axis which is fixed, and you rotate about the X axis, so here is X Y Z,

here is your X axis. So, you rotate like this fate. So, if you do that, then you think of the yz

plane and on the yz plane, you will get a cosine theta minus sine theta sine theta cosine theta and

then it will be a block diagonal with one corresponding to the X Y X has been fixed.

And then for the Y axis, similarly, you have a one in the middle position and the

minor corresponding that one or the matrix corresponding to that one is cosine theta

minus sine theta sine theta cosine theta. So, I will encourage you to check this.

And again, notice again, because of the previous slides, that Ti theta transpose is Ti minus

theta and Ti theta transpose Ti theta is Ti theta transpose is identity for the same reason that we

gave in the, R2 example. Let us do another example. So,

this is a bit more involved to look at the linear transformation, T from R3 to R3, given by X1, X2,

X3 is one third X1 minus 2X2 plus 2X3 2X1 minus X2 minus 2X3 and 2X1 plus 2X2 plus X3. Now, these coefficients should ring a bell in your mind. But if they do not, we will anyway study

what happens to the standard bases vector and you will see that these are very familiar.

So, if you evaluate e1, let us call that V1, then you get one third,

two thirds and two thirds. If you evaluate e2 let us call that V2, you get minus two thirds

minus one third, two thirds. And if you evaluate e3, then you get V3 which

is two thirds minus two thirds and one third. So, we have seen earlier that this

is an ortho normal basis, this was exactly the an orthonormal basis we produced in the

gram Schmidt process video. So, the corresponding matrix is

one third times 1 minus 2 2, 2 minus 1 minus 2, 2 2 1. So, where did this come from? So,

the first column corresponds to V1, the second column corresponds to V2,

and the third column corresponds to V3, and I just taken one third common outside which

I can do, because its scalar multiplication. So, as V1, V2, V3 is an orthonormal set the linear transformation T is an orthogonal transformation. Now, that needs a little bit of checking.

So, what you have to do is, you have to take this T X1, X2, X3 and you have to check that if you

apply the norm on both sides. So, if you take if you take norm of

Tx, then this is the same as the norm of X. This is what you want to check. And the claim is that

if, if I happen to know that V1, V2, V3 is an orthonormal basis, which in this case I do,

then then that will be true. And the reason is this identity

that A transpose is A transpose A is I3 I will suggest you check this identity,

actually it is a very easy check, because what happens is, when you do AA transpose,

or A transpose A, really what you are getting is in each place is the inner product of two of these

basis vectors. So, when they are not equal, it will be 0. And when they are equal, it will be 1.

That is the main sort of reason we said V1, V2, V3 is an orthonormal set.

So, you obtain this and from here you can you can get that norm of Tx is equal to norm of X, because when you apply norm, somewhere in there, this these inner products will play a role.

So, a square matrix is called an orthogonal matrix. If AA transpose is A transpose A is

identity, let me expand a little on that comment. So, the idea is that

when you are looking at norm of Tx, well, you get in the product of Tx comma Tx.

But here we are looking at this in the product in terms of the usual dot product,

so let us do it further dot product. So, Tx dot with Tx, this is the same in

terms of matrices as looking at Tx transpose times Tx, which is the same as x transpose

T transpose T times x, but T transpose T is identities. So, you get x transpose x,

which is norm of x. This is the main point that is why we are demanding this thing here. I may have

used bad notation and probably here, when I went to matrices, I should have

used A and not T just A, every T of x when you evaluate these X as a

vector and R3, as in terms of its coordinates you have to move to A. So, this is why it works. And so, an orthogonal matrix is AA transpose A transpose is identity that is the reason we

call such matrices as orthogonal matrices, because when you use them to create linear transformations

on Rn, then those linear transformations are going to be orthogonal linear transformations.

So, the take home from this video, I guess is that

this notion of orthogonality is it is a very geometric notion, it studies

essentially the angles and in particular right angles which, as we know since antiquity is

something we have been very interested in then there are beautiful theorems related to that.

And this is some linear algebraic way of studying the changes that take place when

you apply a linear transformation to angles and in particular, if they remain unchanged. Then these

are some special linear transformations, and they are called orthogonal linear transformations. Of

course, you have also maintained distances so both are inward. That is it for now. Thank you.